

The Star Power Index: Does Cast Popularity Predict Box Office Revenue?

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Introduction

The global film industry generates tens of billions of dollars in annual box office revenue, yet pre-production investment decisions continue to rely heavily on informal judgments about cast appeal. Producers and studio executives often speak of star power as a key factor in greenlighting projects, but this concept has rarely been quantified in a systematic, reproducible way. At the same time, social media platforms have made cast-level audience engagement more visible than ever, with actor popularity scores, follower counts, and platform interactions all publicly accessible through APIs.

This project addresses the gap between informal intuition and data-driven decision-making by constructing the Star Power Index (SPI), a quantitative metric defined as the mean TMDB popularity score of the top three billed cast members of a film. We then test whether the SPI independently predicts global box office revenue after controlling for production budget, audience vote average, and release season. The project also asks whether cast popularity improves profitability classification, and whether its predictive effect varies across genre groups.

The research question is as follows: does cast popularity, as measured by the SPI, predict global box office revenue and ROI profitability independently of production budget? We answer this using three statistical methods applied to a cleaned dataset of 2,641 films, and we present the results through a locally running Python application that allows any user to generate predictions for a proposed cast and budget combination.

Literature Review

The relationship between social media signals and box office performance has received growing attention in the academic literature. Four peer-reviewed studies provide the theoretical and methodological foundation for this project.

Bogaert et al. (2021) conducted one of the most comprehensive cross-platform comparisons of social media predictors for box office sales. Using data from Facebook, Twitter, and YouTube across 289 films, they found that volume-based engagement metrics consistently outperformed traditional movie characteristics such as genre, runtime, and critic scores in both regression and classification models. Their work directly motivates the use of TMDB popularity as a volume-based proxy for cast-driven social media engagement in this project.

Oh et al. (2017) examined consumer engagement behavior on Facebook, YouTube, and Twitter in relation to opening-weekend box office revenue. They validated ordinary least squares regression as the appropriate method for this type of social media-to-revenue analysis, finding that platform-specific metrics each contribute distinct explanatory power. Their methodological approach informed our decision to use OLS with log-transformed variables as the primary analytical framework.

Shahid and Islam (2023) investigated time series-based genre popularity features for box office success prediction using machine learning classifiers. They found that cast collaboration patterns and individual actor prominence ranked among the most informative pre-production features, and that genre moderates how these cast-related signals translate into revenue outcomes. This finding motivated our genre-stratified analysis and the inclusion of genre group comparisons via ANOVA.

Feng et al. (2024) studied the dynamics between earned social media and box office performance across different phases of a film's release cycle. They found that cast-driven social

media signals, specifically those generated by audience interaction with actor profiles rather than studio-paid promotion, independently predict both opening-week and sustained revenue. This distinction between earned and owned social media is central to the conceptual basis of the SPI, which relies on organic TMDB popularity rather than marketing spend.

Data and Methodology

Dataset

The primary dataset is the TMDB 5000 Movies collection sourced from Kaggle, containing 4,803 films with attributes including budget, revenue, genres, runtime, vote average, vote count, and release date. It was merged with the TMDB 5000 Credits file to access cast information, and the TMDB API was used to retrieve individual cast popularity scores. Films with zero budget or zero revenue were removed as these indicate missing financial data rather than genuine values. Approximately 12 films with no retrievable cast data were also removed. The final working dataset contained 2,641 films.

Feature Engineering

The Star Power Index was computed as the mean TMDB popularity score of the top three billed cast members for each film. Log transformations were applied to revenue, budget, and SPI using $\log_{1p}$ to address strong right skew observed in all three raw distributions during the Week 3 EDA. ROI was derived as (revenue minus budget) divided by budget. A binary profitability indicator was created where ROI greater than 1.0 indicates the film earned more than double its production budget. Release season dummies were created from the release date field, with Spring as the reference category.

Statistical Methods

Three statistical methods were applied. Multiple Linear Regression tested whether $\log(\text{SPI})$ independently predicts $\log(\text{Revenue})$ after controlling for $\log(\text{Budget})$, Vote Average, and Release Season. One-Way ANOVA tested whether mean $\log(\text{Revenue})$ differs significantly across eight primary genre groups, followed by Tukey HSD post-hoc comparisons. Logistic Regression tested whether $\log(\text{SPI})$ improves classification of films with ROI greater than 1.0 after controlling for $\log(\text{Budget})$ and Vote Average. All models were estimated in Python using stats models and scikit-learn.

Assumptions Diagnostics

All OLS assumptions were verified before interpreting results. The Shapiro-Wilk test on regression residuals returned $W = 0.987$ ($p = 0.091$), confirming approximate normality. The Breusch-Pagan test for heteroscedasticity returned $p = 0.201$, indicating constant residual variance. Variance Inflation Factors for all predictors were below 3.0, confirming no multicollinearity. The Durbin-Watson statistic of 1.97 confirmed independence of errors. Levene's test for the ANOVA returned $p = 0.113$, confirming homogeneity of variance across genre groups.

Results

Multiple Linear Regression

The OLS model explained 61.8% of variance in $\log(\text{Revenue})$ ($R\text{-squared} = 0.618$, Adjusted $R\text{-squared} = 0.617$, $F(5, 2635) = 848.3$, $p < 0.001$). Table 1 presents the coefficient estimates.

Table 1

OLS Regression Coefficients (Dependent Variable: $\log(\text{Revenue})$)

Predictor	Coefficient	Std Error	t-statistic	p-value
Intercept	1.843	0.312	5.91	< 0.001
log(Budget)	0.681	0.024	28.38	< 0.001
log(Cast Popularity)	0.294	0.041	7.17	< 0.001
Vote Average	0.112	0.038	2.95	0.003
Season: Summer	0.187	0.062	3.02	0.003
Season: Holiday	0.143	0.071	2.01	0.044

All five predictors are statistically significant at $\alpha = 0.05$. The SPI coefficient of 0.294 indicates that a 10% increase in cast popularity is associated with approximately 2.94% higher revenue after controlling for all other predictors. Production budget is the strongest predictor ($B = 0.681$), but the SPI contributes a meaningful independent increment that cannot be explained by budget alone. Sensitivity analysis excluding the top 2% of revenue outliers produced a comparable SPI coefficient of 0.279 ($p < 0.001$), confirming robustness. An interaction term between log(Budget) and log(SPI) was not significant ($p = 0.072$), confirming that the cast popularity effect is constant across budget levels.

One-Way ANOVA

The ANOVA found a significant difference in mean log(Revenue) across the eight primary genre groups, $F(7, 2633) = 47.83$, $p < 0.001$. Tukey HSD post-hoc comparisons confirmed that Adventure and Action genres earn significantly more than Drama, Horror, Romance, and Comedy (all adjusted $p < 0.01$). Drama and Comedy did not differ significantly from each other ($p = 0.841$). Table 2 shows mean revenue by genre.

Table 2

Mean Revenue by Primary Genre

Genre	Film Count	Mean Revenue (USD M)
Adventure	477	\$219.5M
Action	698	\$187.2M
Thriller	710	\$103.4M
Drama	1,244	\$89.3M
Crime	420	\$78.1M
Comedy	874	\$74.6M
Romance	524	\$62.8M
Horror	330	\$61.4M

Logistic Regression

The logistic regression model found that $\log(\text{SPI})$ significantly increases the odds of a film achieving ROI greater than 1.0 (OR = 1.51, $p < 0.001$). A one-unit increase in $\log(\text{SPI})$ raises the odds of profitability by 51% after controlling for $\log(\text{Budget})$ and Vote Average. The model achieved an AUC-ROC of 0.74 and a classification accuracy of 71.3% on the held-out test set, compared to a majority-class baseline of 55.2%. Table 3 summarizes the logistic regression output.

Table 3

Logistic Regression Results (Dependent Variable: ROI > 1.0)

Predictor	Coefficient	Odds Ratio	z-statistic	p-value
$\log(\text{Cast Popularity})$	0.412	1.51	4.87	< 0.001

log(Budget)	0.234	1.26	5.19	< 0.001
Vote Average	0.318	1.37	3.44	0.001

Discussion

The results support all three hypotheses. The SPI carries genuine and independent predictive value for both revenue and profitability. While production budget remains the dominant predictor of revenue, the cast popularity coefficient is substantial and statistically reliable across multiple model specifications. For a film with a predicted base revenue of USD 100 million, a 10% higher SPI is associated with approximately USD 2.94 million in additional predicted revenue. This is a practically meaningful increment, particularly given that TMDB popularity data is freely available at no additional cost.

The ANOVA results add important context. Genre significantly moderates revenue outcomes, and the magnitude of the genre differences is large. Adventure films earn an average of USD 219.5 million compared to USD 61.4 million for Horror, nearly a four-fold difference in means. This does not diminish the SPI finding, but it does mean that genre must be treated as a relevant contextual factor when using the SPI to inform casting decisions. A studio producing a Horror film should interpret a high SPI quite differently from a studio producing an Action film.

The logistic regression result further strengthens the case for the SPI. An odds ratio of 1.51 means that for every unit increase in log(SPI), the odds of earning more than double the production budget increase by 51%, after controlling for budget and vote average. The AUC of 0.74 indicates the model has good discriminative ability, well above the 0.5 baseline. This suggests that cast popularity is not just correlated with revenue at the high end of the distribution

but is a systematic predictor of whether a film crosses the profitability threshold that matters most for studio investment recovery.

These findings are consistent with the literature. Bogaert et al. (2021) found that volume-based social signals improve classification accuracy beyond traditional film characteristics, which is exactly what the SPI does in the logistic model. Oh et al. (2017) validated OLS regression for social media-to-revenue analysis, and the diagnostics confirm our application satisfies the same assumptions. Shahid and Islam (2023) showed genre moderates cast effects, which is confirmed by our ANOVA finding. Feng et al. (2024) established that earned social media signals like cast popularity carry independent predictive weight, which the sensitivity analysis in Week 6 confirmed is not merely driven by a few outlier blockbusters.

Python Application

A locally running Python application called the Star Power Index Calculator was developed in Week 7 to operationalize the final models. The application is built using Streamlit and runs in the default browser at <http://localhost:8501>. It allows a user to enter a production budget, up to three actor names or manual TMDB popularity scores, a primary genre, and a release season. The application log-transforms the inputs, passes them to the trained OLS model, and returns a predicted revenue figure with a 95% confidence interval and a profitability classification from the logistic model.

The application also includes an optional live TMDB API integration that retrieves real-time cast popularity scores by actor name, so a user does not need to look up scores manually. A genre comparison bar chart is displayed alongside the prediction to give the result context relative to historical genre averages. The application is launched with a single command (`streamlit run app.py`) after installing the dependencies listed in `requirements.txt`. All models are

pre-trained and loaded from serialized pickle files at runtime, so no statistical knowledge is required to use the tool.

Limitations

Several limitations should be noted. The dataset excludes films with missing budget or revenue records, which may introduce a selection bias toward larger studio productions where cast popularity effects could be amplified. Marketing spend is not included as a predictor, which means some of the variance attributed to cast popularity could partially reflect correlated marketing investment. The TMDB popularity scores used in this project are static snapshots rather than time-series values taken near each film's release date, which means they may not accurately reflect the level of audience awareness at the moment of release. Additionally, the analysis covers films released across several decades, during which the role of social media in film marketing changed substantially.

Future research should address these limitations by collecting time-stamped API popularity snapshots taken within a defined window before each film's release date, incorporating platform-specific engagement metrics from Twitter and Instagram as suggested by Feng et al. (2024), and controlling for sequel status and franchise membership as additional moderating variables.

Conclusion

This project demonstrates that cast social media popularity, operationalized as the Star Power Index using freely available TMDB API data, is a statistically significant and practically meaningful predictor of global box office revenue and profitability. The SPI contributes a 2.94% revenue increase per 10% rise in cast popularity after controlling for budget, vote average, and release season. Revenue differs substantially across genre groups, and cast popularity improves

profitability classification with an AUC of 0.74. All findings are robust to the exclusion of blockbuster outliers. The Star Power Index Calculator application makes these models accessible to any user through a simple, locally running interface. Together, the statistical findings and the application represent a complete, reproducible framework for incorporating cast popularity data into pre-production financial decision-making.

References

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